Scalability & Future Plan

Date	01 JULY 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	**Static File-based Data Storage**	Storing raw applications as CSV files on local disk hinders real-time streaming updates and increases disk footprint.	High
2	**Lack of Model Explainability (XAI)**	Rejection decisions from the Random Forest model cannot provide compliance-friendly reasons (e.g., "rejection due to low income"), violating financial regulations.	High
3	**Synchronous In-Memory Inference**	Loading preprocessor and model objects into memory synchronously blocks Flask worker processes during high concurrent prediction traffic.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-threaded Flask development server, running on a free cloud tier with limited concurrent user capacity	Migrate to a containerized Docker setup deployed on AWS ECS or Kubernetes with auto-scaling, run behind a Gunicorn WSGI server.
2	Data Storage	Raw records (application_record.csv and credit_record.csv) read directly from disk via Pandas during asset build.	Port raw data storage to a secure relational database (e.g., PostgreSQL) indexed on applicant ID for fast transactional queries.
3	Performance	Synchronous CPU-bound inference execution for every individual applicant submission	Decouple inference using a Redis cache for duplicate checks and run heavier batch scoring jobs asynchronously using Celery task queues.
4	Security	App data is transmitted over plain HTTP with no applicant authentication or protection of demographic PII.	Implement JWT-based API access control, enforce HTTPS, and encrypt sensitive features (e.g., exact income, age) both at rest and in transit.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	**Explainable AI (XAI) & validation**	2026 (1-2 Months)	Integrates SHAP/LIME to generate clear reasons for approval/rejection and prevents invalid inputs.
Phase 2	**Database Integration & REST APIs**	2026 (3-4 Months)	Replaces local CSV files with PostgreSQL, enabling automated streaming ingestion and third-party credit check integrations.
Phase 3	**Automated Drift & Retraining Pipeline**	2026(5-6 Months)	Continually monitors data drift in income/age distributions and triggers automated model retraining to maintain high prediction accuracy.